The Golden Parabola Heist

Whodunnit? · Inequalities Name: _____

It was the night of the Riverside Art Museum's annual gala. At exactly 9:00 p.m. the lights flickered out for ten minutes. When they came back on, the pedestal in the middle of the Grand Rotunda was empty: **The Golden Parabola**, the museum's most valuable statue, was gone.

Six people were in the building. Each was seen in a different room, and there are six ways the statue could have been smuggled out. You are the lead detective. Somebody in this building did it, and you need to find out **who** took it, **where** they were, and **how** they got it out.

How to crack the case

1. Your teacher will post **10 clue cards** around the room. Each card has a math problem and four answer choices. Every choice is paired with a clue.
2. Solve the problem on the **Show Your Work** page. Find your answer among the choices and read the clue next to it. If your answer is not there, check your work.
3. Use the clue to cross out suspects, places or methods on the case board below. Write the card number beside what it cleared.
4. After all 10 cards, exactly one suspect, one place and one method are left. Those three boxes name the culprit. If you end up with none, or more than one, a clue went wrong: recheck your math!

The Suspects	The Places	The Methods
☐ Ms. Quill Museum Curator	☐ Grand Rotunda	☐ Hidden in a cello case
☐ Chef Marco Gala Caterer	☐ Sculpture Garden	☐ Rolled up in a rug
☐ Dr. Vega Art Historian	☐ Gift Shop	☐ Packed in a catering cart
☐ Mr. Bishop Security Guard	☐ Rooftop Terrace	☐ Swapped with a replica
☐ Ms. Tanaka Cellist	☐ Café	☐ Tied to a balloon bouquet
☐ Prof. Finch Guest Lecturer	☐ Basement Archive	☐ Slipped into a delivery crate

The culprit is _____ in _____ using _____

Show Your Work

The Golden Parabola Heist · Show every step in the matching box.

Name: _____

1. Choice _____	**2.** Choice _____
3. Choice _____	**4.** Choice _____
5. Choice _____	**6.** Choice _____
7. Choice _____	**8.** Choice _____
9. Choice _____	**10.** Choice _____

Clue #1

The Golden Parabola Heist

Solve for x and write the answer as an inequality (or two joined by **or**).

$$x + 7 < 3$$

A. $x < 10$	Prof. Finch was signing books in the Café, and the line of fans stretched out the door.
B. $x < -4$	Chef Marco was plating desserts in the Café in front of a dozen guests.
C. $x > -4$	Ms. Quill was leading a tour of the Sculpture Garden. Eight guests counted her footsteps.
D. $x < 4$	The balloon bouquet by the entrance was popped and checked. Nothing was tied to it, so the statue never floated out that way.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #2

The Golden Parabola Heist

Solve for x and write the answer as an inequality (or two joined by **or**).

$$-4x \geq 20$$

A. $x \leq 5$	Chef Marco was setting out a snack tray in the Sculpture Garden. Guests grabbed mini quiches all through the blackout.
B. $x \geq 5$	Party planners say every gala balloon was tied on with ribbon only. No statue hung from a bouquet.
C. $x \geq -5$	Prof. Finch was giving a mini-lecture in the Grand Rotunda. The audience clapped at 9:02 p.m.
D. $x \leq -5$	The catering cart was searched at the door. No statue inside, so the Parabola was not packed in a catering cart.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #3

The Golden Parabola Heist

Solve for x and write the answer as an inequality (or two joined by **or**).

$$2x - 5 > 9$$

A. $x > 14$	Dr. Vega was sketching a bronze fox in the Sculpture Garden. Her sketchbook page is dated 9:05 p.m.
B. $x > 2$	Prof. Finch was haggling over a poster in the Gift Shop. The cashier remembers the argument.
C. $x < 7$	The balloon bouquet was far too light and wobbly to carry anything heavy out the door.
D. $x > 7$	Dr. Vega was studying old sketches in the Basement Archive. The sign-in sheet has her name all evening.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #4

The Golden Parabola Heist

Solve for x and write the answer as an inequality (or two joined by **or**).

$$\frac{x}{3} + 2 \leq 5$$

A. $x \geq 9$	A guest photo of the balloon bouquet at 9:10 p.m. shows only ribbons and confetti.
B. $x \leq 3$	Mr. Bishop was walking his patrol route through the Sculpture Garden. The checkpoint scanner beeped at 9:01 p.m.
C. $x \leq 9$	A guard checked every cello case at the exit. None held the Parabola.
D. $x \leq 21$	Prof. Finch was reading dusty ledgers in the Basement Archive. The sign-in sheet shows his signature.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #5

The Golden Parabola Heist

Solve for x and write the answer as an inequality (or two joined by **or**).

$$3(x - 2) \geq 2x - 1$$

A. $x \geq 5$	Ms. Quill was giving the welcome speech in the Grand Rotunda. Every guest saw her.
B. $x \leq 5$	The helium tank log shows all balloons were filled before the gala. No bouquet was added later.
C. $x \geq -7$	Ms. Tanaka was tuning her cello among the statues in the Sculpture Garden. Two guests stopped to listen.
D. $x \geq 1$	Prof. Finch was pointing out constellations on the Rooftop Terrace to a group of guests.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #6

The Golden Parabola Heist

Solve for x and write the answer as an inequality (or two joined by **or**).

$$5 - 2x < 11$$

A. $x > 3$	A photographer caught Ms. Quill posing with a donor in the Sculpture Garden right as the lights went out.
B. $x < 3$	Staff untied every balloon bouquet and found nothing hidden in the ribbons.
C. $x > -3$	The rug cleaners confirm nothing was rolled up in any gallery rug.
D. $x < -3$	A waiter swears Prof. Finch was stirring sugar into his coffee at the Café the whole time the lights were out.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #7

The Golden Parabola Heist

Solve for x and write the answer as an inequality (or two joined by **or**).

$$-3 < 2x + 1 \leq 9$$

A. $-2 < x \leq 4$	Mr. Bishop was checking the cameras on the Rooftop Terrace. The video shows him the whole time.
B. $-1 < x \leq 5$	Mr. Bishop locked the gate of the Sculpture Garden at 9:00 and stayed beside it. The gate log backs him up.
C. $-2 \leq x < 4$	The florist who made the balloon bouquets says they could never hold up a statue.
D. $-4 < x \leq 8$	Prof. Finch was trying on a museum scarf in the Gift Shop. A photo from 9:03 p.m. proves it.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #8

The Golden Parabola Heist

Solve for x and write the answer as an inequality (or two joined by **or**).

$$-8 \leq -2x < 6$$

A. $-3 < x < 4$	Security cameras show the balloon bouquet bobbing in the lobby all night. Nothing was tied to it.
B. $-3 \leq x < 4$	Chef Marco was lighting little lanterns along the paths of the Sculpture Garden when the power cut.
C. $-3 < x \leq 4$	The “replica” on display is just the old plaster model, so no swap took place.
D. $-4 \leq x < 3$	Prof. Finch was answering questions by the fountain in the Grand Rotunda with a crowd around him.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

Clue #9

The Golden Parabola Heist

Solve for x and write the answer as an inequality (or two joined by **or**).

$$x - 4 < -3 \quad \textbf{or} \quad 2x \geq 10$$

A. $x < -7$ or $x \geq 5$	Prof. Finch was helping an intern sort boxes in the Basement Archive. The intern vouches for him.
B. $x > 1$ or $x \leq 5$	A kid popped the whole balloon bouquet at the party. No statue was attached.
C. $x < 1$ or $x \geq 20$	Ms. Tanaka was playing a quiet tune in the Sculpture Garden. A dozen guests can hum it back.
D. $x < 1$ or $x \geq 5$	Ms. Tanaka was buying postcards in the Gift Shop. The receipt is time-stamped 9:04 p.m.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.

✂ cut here

Clue #10

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A taxi charges a \$4 fee plus \$3 per mile. Ana has **at most** \$25 to spend. Write and solve an inequality for the number of miles, x, she can ride.

$$4 + 3x \leq 25$$

A. $x < 7$	Prof. Finch was taking selfies on the Rooftop Terrace. The photo timestamps all read between 9:00 and 9:10 p.m.
B. $x \leq 7$	Dock records show the delivery crate was empty, so the statue was never slipped into it.
C. $x \leq 21$	Dr. Vega was explaining a marble fountain to a student group in the Sculpture Garden.
D. $x \geq 7$	The Parabola weighs far more than any balloon bouquet could lift, so it did not leave that way.

Solve the problem, find your answer, and read its clue. Cross it off on your case board.